New ECDLP index-calculus leads, 2026-11-11

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Owner’s request: “go and find more leads”. Three idea-generator agents each took one lane and wrote the files below. I read every file, re-checked the claims each top lead depends on against the code and data on disk, and ranked the leads. All statements are `[pcc:unverified]`. Nothing in this directory has been run, apart from the re-checks listed in “Checks made for this report”. Nothing concerns a deployed curve or key recovery.

Lane	File	Leads
A — last fall degree and the selector proof system	`lane_A_falldegree.md`	5
B — binary-field IC engineering	`lane_B_binary_ic.md`	4
C — prime and extension fields, covers	`lane_C_prime_ext_cover.md`	4

Novelty check. Every lead was compared against the 173 lead ids of the 2026-11-10 sweep (`EXISTING_LEADS.txt`, from `research/lead_sweep_20261110/`). Each lane file names the nearest existing lead and the difference.

Two limits on that check: - The crypto-autoresearcher lead atlas is vault-encrypted (`VLT1`) and was not read. - The `research-kb` MCP server was not available.

So “new” means new with respect to this repository and the sweep only.

Ranked list

#	Lead	Lane	What it would change	Cheapest test	Prior / status
1	`LEAD-C-jv-nagao-hilbert-at-scale`	C	Nagao arm of the Joux–Vitse cover at p = 1301–1601	9 counted runs, about 5 min each, against the frozen rho reference	Measured 1.85–1.88× lower counted S at equal relations, p = 53–503, classified engineering (`research/ic_cover_weakclass_20261023/REPORT.md`); never run at scale

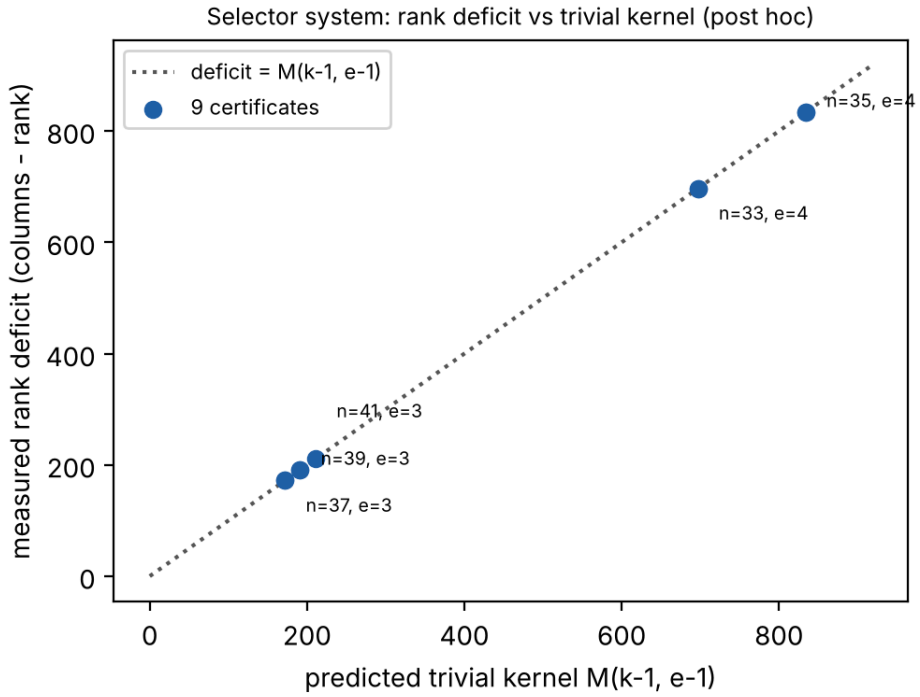
#	Lead	Lane	What it would change	Cheapest test	Prior / status
2	`LEAD-A-selector-eta-kernel`	A	Explains the selector system's rank, the k = 13 anomaly, and a reduction to the trace hyperplane	nullity mode in `emin.c`, under 15 min	Proved core; rank prediction already matches 9 of 9 certificates (below)
3	`LEAD-C-jv-sieve-emit-price`	C	Turns the one validated below-rho row (JV sieve, p = 1511, 0.0065×) from accounting into a priced result (treatise problem 11)	16 cover seeds, under 5 min	Engineering; predicts emitted/proxy 0.80–1.05
4	`LEAD-C-jv-sieve-hilbert-descent`	C	JV sieve total S × about 0.87 at p = 1511	desk check on the 16 records, then a code change	Wiring gap confirmed (below)
5	`LEAD-ICB-d4bin-chain-variable-sweep`	B	k = 4 characteristic-2 per-trial cost (charpoly 45.8 % of the profile)	seconds for step 1	Prior about 10 %
6	`LEAD-A-selector-counting-law`	A	Predicts where `e_min` dips; the Koszul-plus-hyperplane model puts the next dips at k = 16 and 19	small-k grid, under 10 min	Conjecture

#	Lead	Lane	What it would change	Cheapest test	Prior / status
7	`LEAD-A-fall-effective-variable-law`	A	Rule $1 \in W_4$ iff $v \le 33$; explains R33 falls versus T33 stalls	needs the native closure (vaulted; not on disk here)	Fitted rule; may point *against* a growing bound
8	`LEAD-A-closure-dimension-laws`	A	Closed forms for $\dim(W_3 \cap B_{\le 3})$ and $\dim(W_4 \cap B_{\le 3})$	existing closure data	Fitted
9	`LEAD-A-bilinear-lift-size-degree`	A	k^2 extension variables $y_{ij} = a_i c_j$; claims $d_F\ 4 \to 3$ at $n = 15$	small closure, minutes	Instance of the size–degree trade-off of NOTE.md Lemma 2
10	`LEAD-ICB-tableB-binary-known-log-claw`	B	Table B random-binary rows: predicts 1.4–2.4× matched rho, against 10.6–38.2× today	existing `ic.known_log_mitm2_v1` arm, under 5 min	**Needs an owner ruling first** (below)
11	`LEAD-C-quartic-n4v2-k26`	C	Plane-quartic double-large-prime run from $k = 22$ to $k = 26$	needs 128-bit group-order arithmetic	Extends an existing row
12	`LEAD-ICB-d4bin-constant-quadratic-elimination`	B	$k = 4$ elimination of the weight-2 unknowns	about a day to implement	The agent's own count predicts *worse*; null registered

#	Lead	Lane	What it would change	Cheapest test	Prior / status
13	`LEAD-ICB-d4bin-exponent-coset`	B	Factor 2–4 on k = 4 if the 21 carrying monomials lie in an index-2 coset	one-minute probe	Prior about 5 %

Why the top two. - **Lead 1** is the cheapest test with the largest predicted *measured* improvement of an existing IC row. Its mechanism (Hilbert early stop of F4) is already validated at smaller p, and the scale runs simply never used it. - **Lead 2** has a proved core and a sharp falsifier: a non-trivial c-only syzygy. Data from this morning's runs already agree with it.

Leads 1, 3 and 4 move IC rows. Leads 2 and 6–9 bear on treatise problems 12 and 13, not on any row.



Checks made for this report

- **Lead 1, scale runs never used the Hilbert stop. Holds.** `research/jv_cover_crossover_20261101/run.sh`, `jv_cover_fbline_20261102/run.sh` and `jv_cover_p_scaling_20261024/run.sh` call `--exp dlp` with no `--stop`. The Hilbert rule was measured only in `research/f4_early_stop_20261018/`.
- **Lead 1, the 1.85× figure. Holds.** It comes from `research/ic_cover_weakclass_20261023/REPORT.md`: arm B (`hilbert:16`) lowers counted S by 1.848–1.875 per cell at p = 53–503, with equal relations in 25 of 25 cells. That report classifies it as engineering, not an advance.
- **Lead 4, the sieve's descent cannot use the Hilbert stop. Holds.** `src/cryptanalysis/jv_sieve.rs:2086` builds its F4 options with `opts_for(24, 600.0, stop)`. In `src/cryptanalysis/jv_cover.rs:2734`, `opts_for` takes `Option<usize>` and only selects the staircase stop. The Hilbert rule is reachable only through `opts_for_mode` (`jv_cover.rs:2726`), which the sieve does not call.
- **Lead 5, the chain variable is fixed. Holds in substance.** `src/cryptanalysis/gaudry_quartic.rs:2235` takes `mvar` as the first variable of weight 1, so the choice is fixed by the weight vector and never searched.
- **Lead 2, rank deficit equals `M(k−1, e−1)`. Matches 9 of 9.**
  - Claim: the rank deficit of the selector system equals `M(k−1, e−1) = sum_{d<=e−1} C(k−1, d)`, the trivial homogeneous kernel.
  - Data: the coefficient-matrix ranks of every independently re-checked infeasibility certificate in `research/lfd_lower_bound_20261111/out/c/verify_*.txt`.
  - Result: deficit and prediction agree exactly in all 9 cases, at n = 33, 35, 37, 39, 41 and e = 3, 4 (for example n = 35, e = 4: 141,680 − 140,846 = 834 = M(17, 3)). Figure: `figures/deficit_vs_kernel.svg`.
  - Caveat: the agent formed the claim from these same logs, so the match is a post-hoc consistency check, not the preregistered test. The test is the nullity mode the lead names.
- **Lead 10 needs an owner ruling before any work.** `ic.known_log_mitm2_v1` is a BSGS-type (known-log table) arm. The owner ruled on 2026-11-08 that BSGS-class results are labelled boundary results outside the index-calculus ratchet (`research/curve25519_autolab_20261104/bsgs/README.md`). As things stand, a win for lead 10 would not count as an IC improvement.

What none of these is

None is a growing lower bound on d_F, and none is a sub-√p claim.

No lead is ready for an IC leaderboard row. Each would first have to pass its test, then the validation gate (replay certificate, validator report, matched rho with the §5a band).

Leads 1, 3, 4 and 11 should run on Iso Lab EKS once an AWS identity works. Until then, local runs are development checks only.